MIL-T-62051C(AT)
2 May 1986
SUPERSEDING
MIL-T-62051B(AT)
10 June 1980

MILITARY SPECIFICATION

TRUCKS, TANK: PROPANE DELIVERY; DIESEL, GASOLINE AND LIQUEFIED PETROLEUM GAS ENGINE DRIVEN; 2,400 GALLON WATER CAPACITY, 24,000 POUNDS GVW, 4x2, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel, gasoline, and liquefied petroleum gas engine driven, four-wheel, two-rear-wheel drive, commercial, 2,400 gallon minimum water capacity, liquid propane delivery trucks having a minimum gross vehicle weight (GVW) of 24,000 pounds, for normal operating conditions. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

O-E-915	- Extinguisher, Fire, Dry Chemical (Portable).
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Specification Comment Sheet appearing at the end of this document or by letter.

AMSC N/A FSC-2320

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

* ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway)

STANDARDS
FEDERAL
FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
MILITARY
* MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Dangerous Articles in Tank Motor Vehicles.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL BUREAU OF STANDARDS
Handbook 44 - Specifications and Regulations for Commercial Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

* (Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted, shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, 345 East Forty-Seventh Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA No. 58 (1983 edition) - Standard for Storage and Handling, Liquefied Petroleum Gases.
NFPA No. 407 (1985 edition) - Aircraft Fuel Servicing.

(Application for copies of NFPA publications should be addressed to National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, INC.
SAE Standards and Recommended Practices.

	J245	- Engine Rating Code - Spark Ignition.
*	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
*	J537	- Storage Batteries (DoD adopted).
*	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
*	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
*	J688	- Truck Ability Prediction Procedure (DoD adopted).
*	J704	- Transmission Mounted Power Take-Offs (DoD adopted).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

J816 - Engine Test Code - Spark Ignition and Diesel.

* J1349 - Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2), except the upper two thirds of the tank shall be painted gloss white.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions and rear axles shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. When specified (see 6.2), a heavy duty cooling system shall be furnished that will maintain the engine coolant at a temperature below the boiling point with vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.5 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2-quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel. On tilt cab models, a radiator servicing access door shall be provided to allow verification of coolant level.

3.1.1.6 Towing devices. Not less than two hooks, loops, or pins for towing the vehicle shall be furnished on front of the vehicle.

* 3.1.1.7 Wheel splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. Vehicles with tilt cabs shall have rubber mud flaps to the rear of the front wheels. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682. A metal strip not less than 1/8-inch thick and not less than one inch wide, extending the entire width of the mud flap shall be installed to preclude bolt heads or the bolt nuts from damaging the mud flaps.

3.1.1.8 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

* 3.1.1.9 Bonding straps. Bonding straps shall be attached using corrosion resistant bolts and nuts and using two internal-external locking washers at each attaching point to assure positive electrical connection. Bonding straps shall be furnished between the chassis frame and the cab, fenders, engine hood, and tank body, and also between the pump, meter or other static producing devices and the nearest structural part of the body or chassis.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body; and full complement of chassis fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The distribution of gross vehicle weight for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity. Minimum GVW rating shall be 24,000 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

* 3.2.8 Accessibility. The design of the vehicle and optional equipment shall allow access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a continuous grade of 1.0 percent at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive and, when a multi-speed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed and shall provide not more than the vehicle speed (mph) specified in table I for the corresponding number of available forward speeds.

TABLE I. Low speed requirement.

Number of forward speeds	Speed (mph)
5	5.0
10	3.0

* 3.3.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Engine governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.1 Engine. The engine furnished for the specified vehicle shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2 and 3.4.1.3), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J245 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.3 Liquefied petroleum gas (LPG) engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle liquefied petroleum gas (LPG) conversion engine with not less than 6 cylinders. The engine shall produce the required performance when operated on LPG, at engine speed not more than the manufacturer's recommended operating speed. Internal engine modifications shall include valve seat inserts, hardened intake valves, stellite faced exhaust valves, heat block for carburetor and pistons providing a higher compression ratio. Conversion shall include replacement of carburetor with an LPG carburetor, addition of a pressure regulator with coolant lines, addition of an LPG fuel tank and fuel lines, addition of an LPG magnetic fuel lock filter in place of gasoline type, modification of air cleaner warm air devices for LPG requirements, replacement of fuel pump with a cover over pump opening, readjustment of spark advance and installation of cold-weather starting aids. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.4 Oil filter. A full flow type oil filter shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.1.5 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. For diesel engine driven vehicles, control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), coolant heater; engine oil heater; and fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heaters, 1,500-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed.

(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

(d) A fuel warmer or preheater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt starting system, with 12-volt direct current (dc) lighting system shall be furnished. A minimum 60-ampere alternator shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type ether systems shall be key operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system. For gasoline and LPG engine driven vehicles, a 12-volt dc ignition system shall be furnished. An alternator of not less than 40 amperes rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with double-faced front signal units and single-faced rear signal units installed in accordance with SAE J588 except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall have a visible and audible type flash indicator.

* 3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and total cold cranking ampere rating

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

at 0°F, both measured in accordance with SAE J537, shall not be less than specified in table II. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE II. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1200
Gasoline and LPG	110	450

* 3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67. LPG engine driven vehicles shall conform to NFPA No. 58, Section 3-6.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than 1 quart.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) of not less than 50 gallons total capacity (43 gallons for vehicles destined for California) shall be furnished for gasoline and diesel engine driven vehicles. When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage. Fuel capacity for LPG engine driven vehicles shall provide a cruising range of not less than 200 miles.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. If vertical exhaust mufflers are furnished and if they are capable of being reached easily by personnel entering or leaving either side of the cab, a heat shield shall be provided. Vertical exhaust systems, if furnished, shall be provided with hinged rain caps. The exhaust system shall comply with NFPA No. 407. A spark arrester shall be furnished, except on turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. In addition, the vehicle shall conform to 3.4.4.1 or 3.4.4.2, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.4.1 Diesel engine exhaust. The tail pipe shall be stacked at the rear of the cab.

* 3.4.4.2 Gasoline and LPG engine exhaust. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83 and paragraph 4-3.4 of NFPA No. 407. The exhaust outlet shall be located immediately behind the cab (between the cab and the tank body) or on the side immediately in front of or to the rear of the rear wheels. The exhaust outlet, if to the side, shall be on the side opposite the dispensing compartment. All portions of the exhaust system which project to the rear of the cab shall be fully shielded from fuel sources. The shield(s) shall direct fluids spilled from the tank, leaking from the tank, or spraying from any leaking seals, away from the exhaust system. Shields shall be separated and not less than 1-1/2 inches, measured vertically, and not less than 6 inches, measured horizontally, from the nearest exhaust system component. The shielding shall not interfere with servicing any component under the tank body requiring periodic servicing.

3.4.5 Transmission. Input torque capacity shall be at least equal to the maximum torque delivered by the engine.

* 3.4.5.1 Manual transmission. Unless otherwise specified (see 3.4.5.2), a manually shifted transmission with a minimum of five forward speeds shall be provided. When specified (see 3.4.9.1), a two-speed rear axle shall be furnished. Gear ratios in transmissions and axles shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide required vehicle performance (see 3.3.1). The transmission shall be provided with SAE J704 power takeoff openings at right and left sides. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.1.1 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle and engine furnished with clutch torque capacity exceeding maximum delivered engine torque.

* 3.4.5.2 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff opening(s). The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

* 3.4.5.3 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Driveshaft shall be protected in a manner similar to that specified for buses by Federal Motor Carrier Safety Regulation 393.89.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the vehicle furnished. Frame rails shall not project beyond the rear end of the body.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to specified GVW. When suspension is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. Unless a two-speed axle is furnished, a single reduction rear axle shall be provided.

3.4.9.1 Two-speed axle. When specified (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper transmission gear splitting. Gear ratio shall provide performance specified in 3.3.1.

3.4.9.2 Traction control. When specified (see 6.2), traction control shall be furnished on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rim and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:22Z
Check the source to verify that this is the currenversion before use.

shall be the same for all tires on the vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type tires and wheels for the front and rear axle in lieu of duals. Wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle loaded to specified GVW. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. For tube type tires, inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of spare wheel assembly using only the tools specified in 3.4.16.1.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes. The braking system, complete with all necessary components, shall include:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

(a) Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cubic feet per minute (cfm).
(b) Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
(c) Foot control, suspended or treadle-type.
(d) Air control valves.
(e) Air pressure gage, visible to the driver.
(f) Low-air-pressure warning, visible and audible.
(g) Service brake stop lamp switch.
(h) Alcohol aspirator, with unbreakable container.
(i) Automatic moisture ejector.
* (j) Automatic slack adjusters.
* (k) When specified (see 6.2), in lieu of subparagraph (h), an air dryer shall be furnished. The air dryer shall be of the desiccant or after-cooled type.

3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full-width cab. When specified (see 6.2), a tilt type cab, with tilting and locking mechanism, shall be furnished. Tilt cab features shall be in accordance with Employee Safety and Health Standards of Federal Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. The cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be installed. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield. Safety grips shall be provided on each side of the cab to assist personnel to enter or leave cab. A "NO SMOKING" sign visible to seated personnel shall be furnished.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.15 Bumpers. Unless the front bumper is an integral part of the vehicle cab, the bumper shall be of the channel type.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. The rear bumper shall protect the tank, piping, valves and fittings.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechansims shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch.
(b) Ammeter, charging indicator or voltmeter.
(c) Fuel gage.
(d) Oil pressure gage or indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

(e) Engine temperature gage or indicator.
(f) Speedometer with recording odometer.
(g) Dual sunvisors.
(h) Front door mounted armrest on driver's side.
(i) Driver's compartment ventilator other than window.
* (j) Tachometer
* (k) No cigarette lighter shall be furnished.
* (l) An ash tray, if furnished, shall be secured shut.

3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas inclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors on conventional cabs shall have not less than three supporting arms. On tilt cabs, the mirrors shall have not less than two supports.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

* 3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of chassis engine operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Backup alarm. When specified (see 6.2), an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when vehicle is in reverse gear operation, shall be furnished.

* 3.5 Body and equipment. The vehicle shall be equipped with a 2,400 gallon water capacity tank for the transportation and delivery of liquid propane. The tank shall be mounted on a chassis having a cab-to-axle dimension as recommended by the tank manufacturer. The unit including all operating equipment shall conform to 49 CFR section 178.337, Specification MC 331 as published in the latest issues of the Code of Federal Regulations and as modified by all issues of the Federal Register published and in effect prior to completion of the vehicle. The unit shall also conform to chapter 6 of NFPA No. 58. The owner's copy of sketches and certifications as required by MC 331 and ASME Section 8 shall be shipped with the vehicle. Design pressure and materials used throughout the unit shall be as required by MC 331 for at

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

least 250 pounds per square inch gage (psig) working pressure. The fuel delivery system shall have a capacity of 90 gallons per minute (gpm) at the discharge end of the pump. The complete unit shall include accessories, safety devices and controls as furnished to the general public for the delivery of propane. Dissimilar metals shall not be used in intimate contact with each other. The tank and all associated fittings, components, and piping shall be free of all dirt, loose scale, and petroleum soluble coating.

* 3.5.1 Tank. The tank shall be designed, fabricated, tested, inspected and labeled as specified in the ASME Boiler and Pressure Vessel Code, Section 8. The length of the tank shall provide load distribution proportional to the rated capacity of the front and rear axles of the chassis. The tank shall be cylindrical with one piece hemispherical front and rear heads. The tank shall have a single 2,400 gallon compartment. Baffles shall be provided, as required, to reduce the length from baffle to baffle and baffle to head to not more than 90 inches. Each baffle shall be provided with an 18-inch diameter manway opening.

3.5.1.1 Manholes. A round or oval propane type manhole opening not less than 15 inches in diameter shall be furnished.

* 3.5.1.2 Welding procedures, welders and weld testing. All welding procedures for the tank, and tank attachments shall be qualified in accordance with ASME Section IX and shall be current as specified in CFR title 49, 178.337. Non-destructive examination of production welds for the tank attachments shall be as specified in ASME Section 8. Welders shall be qualified in accordance with ASME Section IX. Welder qualification coupons shall be identified and made available to the Government representative for review (see 4.2.1 and 4.2.2.).

* 3.5.1.3 Weld workmanship specimens. Workmanship specimens of all tank welds shall be prepared by the contractor or manufacturer to duplicate all conditions under construction. Specimens shall represent the minimum acceptable quality levels to be used in production subject to approval by the government (not to include ASME Section 8 welds).

3.5.2 Pump and power takeoff. A positive displacement rotary propane pump shall be furnished and driven from a power takeoff attached to the truck transmission. The power takeoff shall be operable from the cab and by remote control from the rear platform (see 3.5.7). Both cab and remote hand throttle controls shall be furnished and permanently set to limit engine rpm to speed required for a rated flow capacity of the propane pump. The pump shall provide the performance specified in 3.5 at a power takeoff shaft speed of 500 to 700 rpm. The pump shall have 2-1/2-inch minimum suction and discharge openings and a built-in safety relief valve. The pump shall be of the flanged type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.5.3 Intake and discharge system. The intake or discharge system shall incorporate the following valves and equipment functionally arranged:

(a) Flexible connectors.
(b) Vapor eliminator.
(c) Meter, 2-inch minimum.
(d) Differential regulator.
(e) Manual shut-off valves.
(f) Bypass valve (external to pump).
(g) Tank outlet valve control.
(h) Hydrostatic relief valve.
(i) Thermometer.
(j) Pressure gage with valve.
(k) Magnetic float gage with 4-inch dial.
(l) Pressure gage.
(m) Back check valves.
(n) Vapor equalizing valve.
(o) Filler valve.

3.5.3.1 Spray bar loader. When specified (see 6.2), where fast loading of tank will be necessary, a nominal 10-foot long spray bar loader shall be furnished.

3.5.4 Piping. All piping shall be not less than 2-1/2-inch nominal inside diameter, with long sweep bends and shall be installed to resist expansion and vibration without failure. A 2-1/2-inch line strainer with brass screen shall be installed forward of the pump.

3.5.5 Measuring meter. A positive displacement meter, of a capacity recommended by the manufacturer and calibrated for handling commercial propane, shall be provided. This meter shall have a minimum 2-inch inlet and outlet and shall be accurate within one percent at 50 gpm flow rate and accurate within two percent at flow rates between 35 and 100 gpm. The meter shall have a seven figure totalizer to record the total gallons pumped and a four figure set-back counter with figures not less than 3/4 inch high. The meter shall be located to facilitate reading of numerals and shall conform to NBS H-44.

3.5.6 Hoses and hose reels. Not less than 100 feet of 1-1/4-inch inside diameter LPG delivery hose and 100 feet of 3/4-inch LPG vapor hose shall be furnished. The hoses shall be furnished with valves and adapters and mounted on electric rewind hose reels. Hose reel motors shall be provided with circuit breakers. Rollers shall be provided, when necessary, to prevent damage to hose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.5.7 Rear platform. A rear platform shall be provided. The rear platform shall incorporate remote controls for unit and power takeoff operations, truck lighting, tool box(es), hose reels, aluminum or magnesium wheel chocks (one each side), fire extinguisher (see 3.5.7.1), and rear end protection (see 3.4.15.1).

3.5.7.1 Fire extinguisher. A fire extinguisher conforming to O-E-915 size 20, type I or II, class 2 shall be provided. The fire extinguisher shall be mounted in a readily accessible location on the rear platform.

3.5.8 Rear fenders. Sturdy metal fenders for the rear wheels shall be provided.

3.5.9 Instruction plate. A metal plate showing a schematic layout of all valves, emergency valve controls and equipment with full tank operating instructions shall be mounted in a readily visible location at the rear of the vehicle. The plate shall be fabricated to withstand extreme weather conditions. Identification plates, attached to valves, strainers, pump, meter, and other related items, shall be masked prior to painting.

3.5.10 Lights and wiring. In addition to the lighting specified in 3.4.2.3 and 3.4.2.4, the lighting and wiring requirements are as follows:

(a) A light shall be furnished to illuminate the meter.
(b) All body wiring shall be in conduit with threaded joints, having waterproof and vaporproof connections at the lamps and junction boxes, but need not include boxes around lights.

3.5.11 Dust cap. Each service outlet shall be provided with a dust cap, complete with security chain.

3.5.12 Tool box(es). Tool box(es) shall be provided for storage of all accessories normally detachable.

3.5.13 Ease of maintenance. All valves, meters, floats, and those accessories requiring repair, replacement or lubrication shall be installed with removable fittings in the most practical manner for ease of maintenance.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: focusing of lights; adjustment of the engine, electrical and brake systems; filling and charging of battery; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.2.1 Welders qualification records. All welders used in production welds of the tank and tank attachments shall be tested in accordance with ASME Section IX. Welders coupons/or radiographs shall be uniquely identified (see 3.5.1.2 and 3.5.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

* 4.2.2 In-process inspection. The first production tank truck being produced under the contract shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be to evaluate workmanship, weld specimens, and procedures (see 3.5.12 and 3.5.13). The contractor shall be prepared to furnish the government with inspection documentation records verifying that all testing required under ASME Section 8 for the type of tank construction provided, has been accomplished (for example, material test reports, impact test, heat treat charts, hydrostatic test, data reports, et cetera).

4.3 Final cleaning. The tank shall be inspected for cleanliness to determine conformance to 3.5.

* 4.4 Availability of regulations. The contractor shall make available to the Government at the point of final acceptance a current copy of CFR title 49 178.337, Specification MC 331 and applicable portions of the ASME Code. The documents shall be surrendered to the Government for the personal use of Government representatives in conjunction with the contract for the duration of the contract.

* 4.5 Certification. A manufacturer's certificate of compliance with MC 331 as defined in 178.337 and ASME Section 8 shall be furnished, endorsed by the vehicle contractor if different, and securely stowed and shipped with the vehicle together with instructions for the recipient regarding the certificate's retention and use. The certificate shall bear, in addition to the signatures specified above, the typed and signed name of the Government official accepting the vehicle.

4.6 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract and the design adequacy of the vehicle. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.6.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles shall be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds, and the payload (see 3.2.5) to provide the specified GVW. Calculated imposed loads on front and rear axles shall be compared to the suspension, axle and tire load ratings to determine if these components are of adequate capacity to meet contract requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

* 4.6.2 Road test. The first production vehicle (with payload) shall be examined and road tested by the contractor for a distance of not less than 25 miles, at speeds up to 50 mph, to assure that the vehicle will operate in accordance with the contract requirements.

* 4.6.3 Truck body treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.6.4 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

4.6.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.6.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.7 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. The supplier shall maintain records of the non-destructive examinations on all welds. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment from the manufacturer's plant to the initial receiving activity in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for general administrative use by the Government in transporting and delivering liquid propane by personnel trained and instructed in the handling of propane.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Concealed marking, if required (see 3.1.1.1).
(d) Rustproofing, if required (see 3.1.1.2).
(e) Tropical rustproofing, if required (see 3.1.1.2).
(f) Heavy duty cooling system, if required (see 3.1.1.4).
(g) Gasoline engine, if required (see 3.4.1.2).
(h) Liquefied petroleum gas engine, if required (see 3.4.1.3).
* (i) Power plant heaters and fuel warmer, if required (see 3.4.1.7).
* (j) If a battery heater is not required (see 3.4.1.7).
(k) Automatic or semiautomatic transmission, if required (see 3.4.5.2).
(l) Two-speed axle, if required (see 3.4.9.1).
(m) Traction control, if required (see 3.4.9.2).
(n) Wide base type rear tires and wheels in lieu of duals, if required (see 3.4.10).
(o) Disc type wheels, if required (see 3.4.10).
(p) Bias ply tires, if required (see 3.4.10.1).
(q) A carrier, if required (see 3.4.10.3).
(r) Spare wheel or rim, if required (see 3.4.10.4).
(s) Spare tire assembly, if required (see 3.4.10.5).
(t) Air dryer, if required (see 3.4.11.1(j)).
(u) Tilt type cab, if required (see 3.4.12).
(v) Tools, if required (see 3.4.16.1).
(w) Air operated horn, in addition, if required (see 3.4.21).
(x) Backup alarm, if required (see 3.4.23).
(y) Spray bar loader, if required (see 3.5.3.1).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirements for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item I should include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Table 2 - Altitude factor. 1.00.
Table 3 - Rolling factor. 1.613.
Table 4 - Area factor. 0.173.
Table 5 - Velocity factor. 250.0.
Table 6 - Altitude factor. 1.00.
Table 7 - Chassis friction horsepower. Use applicable power unit GVW (to nearest, higher 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8 - Grade factor. 0.75.
Table 8A - Correction factor. Not required.
Table 9 - Road factor. 0.0

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - AT

Preparing activity:
Army - AT
Project No. 2320-A278

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.